

Problems based on Properties of analytic function

Theo:- If $f(z)$ is an analytic function with constant modulus show that $f(z)$ is constant.

pf:-

let $f(z) = u + iv$ be analytic function.

given $|f(z)| = c$ (c is constant)

ie $\sqrt{u^2 + v^2} = c$

ie $u^2 + v^2 = c^2$

To show $f(z) = \text{const.}$

• $u^2 + v^2 = c^2$

ie diff w.r.t. x

$$2u \frac{\partial u}{\partial x} + 2v \frac{\partial v}{\partial x} = 0 \Rightarrow 2u u_x + 2v v_x = 0 \quad \text{--- (1)}$$

diff w.r.t. y

$$2u \frac{\partial u}{\partial y} + 2v \frac{\partial v}{\partial y} = 0 \Rightarrow 2u u_y + 2v v_y = 0 \quad \text{--- (2)}$$

$f(z)$ is analytic $\Rightarrow u_x = v_y, u_y = -v_x$

ie

$$2u u_x + 2v v_x = 0 \quad \text{--- (1)}$$

$$-2u v_x + 2v u_x = 0 \quad \text{--- (2)}$$

squaring and adding

$$4u^2 u_x^2 + 4v^2 v_x^2 + \cancel{4u v u_x v_x} + 4u^2 v_x^2 + 4v^2 u_x^2 - \cancel{8u v u_x v_x} = 0$$

$$4u^2 (u_x^2 + v_x^2) + 4v^2 (u_x^2 + v_x^2) = 0$$

$$4(u^2 + v^2) (u_x^2 + v_x^2) = 0$$

$$4c^2 (u_x^2 + v_x^2) = 0$$

$$\text{ie } u_x^2 + v_x^2 = 0$$

$$\text{ie } |f'(z)|^2 = 0$$

$$\text{ie } |f'(z)| = 0$$

$$\Rightarrow f'(z) = 0$$

$$\Rightarrow f(z) = \text{const.}$$

If $f(z)$ is analytic function of z ,
 prove that

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) |f(z)|^2 = 4 |f'(z)|^2$$

$$\nabla^2 [\operatorname{Re}(f(z))]^2 = 2 |f'(z)|^2$$

$$\nabla^2 [\log |f(z)|] = 0$$

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) |f(z)|^p = p^2 |f(z)|^{p-2} |f'(z)|^2$$

Note $\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}$

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) |f(z)|^2 = 4 |f'(z)|^2$$

$$f(z) = u + iv, \quad |f(z)| = \sqrt{u^2 + v^2}$$

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) (u^2 + v^2)$$

$$= \frac{\partial^2}{\partial x^2} (u^2 + v^2) + \frac{\partial^2}{\partial y^2} (u^2 + v^2)$$

$$\frac{\partial}{\partial x} (2u u_x + 2v v_x) + \frac{\partial}{\partial y} (2u u_y + 2v v_y)$$

$$= 2 [u u_{xx} + u_x^2 + v v_{xx} + v_x^2] + 2 [u u_{yy} + u_y^2 + v v_{yy} + v_y^2]$$

$$= 2u [u_{xx} + u_{yy}] + 2v [v_{xx} + v_{yy}] + 2 [u_x^2 + v_x^2 + u_y^2 + v_y^2]$$

$$= 2u(0) + 2v(0) + 2 [u_x^2 + v_x^2 + u_y^2 + v_y^2]$$

• $[f(z)]$ is analytic i.e. u & v are harmonic.

$$= 2[u_x^2 + v_x^2 + u_y^2 + v_y^2]$$

$$u_x = v_y, \quad u_y = -v_x$$

$$= 2[u_x^2 + v_x^2 + v_x^2 + u_x^2]$$

$$= 4[u_x^2 + v_x^2]$$

$$= 4 |f'(z)|^2$$

$$\begin{aligned} f'(z) &= u_x + i v_x \\ |f'(z)| &= \sqrt{u_x^2 + v_x^2} \end{aligned}$$

$$(b) \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) [Re(f(z))]^2$$

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) u^2$$

$$= \frac{\partial^2}{\partial x^2} u^2 + \frac{\partial^2}{\partial y^2} u^2$$

$$= \frac{\partial}{\partial x} [2u u_x] + \frac{\partial}{\partial y} [2u u_y]$$

$$= 2[u u_{xx} + u_x^2 + u u_{yy} + u_y^2]$$

$$= 2u(u_{xx} + u_{yy}) + 2(u_x^2 + u_y^2)$$

$$= 0 + 2(u_x^2 + u_y^2) \quad [\text{as } u_{xx} + u_{yy} = 0]$$

$$u_x = v_y, \quad u_y = -v_x$$

$$= 2(u_x^2 + v_x^2)$$

$$= 2 |f'(z)|^2$$

To prove

$$2 |f'(z)|^2$$

$$= 2(u_x^2 + v_x^2)$$

Q Show that

$$\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} = 4 \frac{\partial^2}{\partial z \partial \bar{z}}$$

Proof:- $x = \frac{z + \bar{z}}{2}, \quad y = \frac{z - \bar{z}}{2i}$

$$\frac{\partial f}{\partial z} = \frac{\partial f}{\partial x} \frac{\partial x}{\partial z} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial z}$$

$$\frac{\partial f}{\partial z} = \frac{\partial f}{\partial x} \cdot \frac{1}{2} + \frac{\partial f}{\partial y} \left(\frac{1}{2i} \right)$$

$$\frac{\partial f}{\partial z} = \frac{1}{2} \left(\frac{\partial}{\partial x} + \frac{1}{i} \frac{\partial}{\partial y} \right) f$$

$$\left[2 \frac{\partial}{\partial z} = \frac{\partial}{\partial x} + \frac{1}{i} \frac{\partial}{\partial y} \right] \text{ --- (1)}$$

Now $\frac{\partial f}{\partial \bar{z}} = \frac{\partial f}{\partial x} \frac{\partial x}{\partial \bar{z}} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial \bar{z}}$

$$\frac{\partial f}{\partial \bar{z}} = \frac{\partial f}{\partial x} \cdot \frac{1}{2} + \frac{\partial f}{\partial y} \left(-\frac{1}{2i} \right)$$

$$\frac{\partial f}{\partial \bar{z}} = \frac{1}{2} \left(\frac{\partial}{\partial x} - \frac{1}{i} \frac{\partial}{\partial y} \right) f$$

$$2 \frac{\partial f}{\partial \bar{z}} = \left(\frac{\partial}{\partial x} - \frac{1}{i} \frac{\partial}{\partial y} \right) f$$

$$\text{or } \left[2 \frac{\partial}{\partial \bar{z}} = \frac{\partial}{\partial x} - \frac{1}{i} \frac{\partial}{\partial y} \right] \text{ --- (2)}$$

$$\text{Now } 4 \left(\frac{\partial}{\partial z} \right) \left(\frac{\partial}{\partial \bar{z}} \right) = \left(\frac{\partial}{\partial x} + \frac{1}{i} \frac{\partial}{\partial y} \right) \left(\frac{\partial}{\partial x} - \frac{1}{i} \frac{\partial}{\partial y} \right)$$

$$\left[4 \frac{\partial^2}{\partial z \partial \bar{z}} = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right]$$

$$\begin{aligned} (a+ib)(a-ib) \\ = a^2 + b^2 \end{aligned}$$

Q:- If $f(z) = u + iv$ is regular function of z in a domain D , then

$$\nabla^2(u^p) = p(p-1)u^{p-2} |f'(z)|^2$$

Soln. $\nabla^2(u^p) = \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) (u^p)$

$$= \frac{\partial^2}{\partial x^2} u^p + \frac{\partial^2}{\partial y^2} u^p$$

$$= \frac{\partial}{\partial x} (p u^{p-1} u_x) + \frac{\partial}{\partial y} (p u^{p-1} u_y)$$

$$= p (u^{p-1} u_{xx} + (p-1) u^{p-2} u_x^2)$$

$$+ p (u^{p-1} u_{yy} + (p-1) u^{p-2} u_y^2)$$

$$= p u^{p-1} (u_{xx} + u_{yy}) + p(p-1) u^{p-2} (u_x^2 + u_y^2)$$

$$= 0 + p(p-1) u^{p-2} |f'(z)|^2$$

$$u_{xx} + u_{yy} = 0$$

$$\begin{aligned} |f'(z)| &= |u_x + i v_x| \\ &= |u_x - i u_y| \\ &= \sqrt{u_x^2 + u_y^2} \end{aligned}$$

Q If $f(z)$ is a holomorphic function of z show that

$$\left\{ \frac{\partial}{\partial x} |f(z)| \right\}^2 + \left\{ \frac{\partial}{\partial y} |f(z)| \right\}^2 = |f'(z)|^2$$

Soln

$$f(z) = u + iv$$

$$|f(z)| = \sqrt{u^2 + v^2}$$

L.O.S

$$\left\{ \frac{\partial}{\partial x} |f(z)| \right\}^2 + \left\{ \frac{\partial}{\partial y} |f(z)| \right\}^2 \quad \text{--- (1)}$$

• taking

$$\begin{aligned} \frac{\partial}{\partial x} |f(z)| &= \frac{\partial}{\partial x} \sqrt{u^2 + v^2} \\ &= \frac{1}{2} (u^2 + v^2)^{-1/2} (2u u_x + 2v v_x) \end{aligned} \quad \text{--- (A)}$$

Similarly;

$$\begin{aligned} \frac{\partial}{\partial y} |f(z)| &= \frac{\partial}{\partial y} \sqrt{u^2 + v^2} \\ &= \frac{1}{2} (u^2 + v^2)^{-1/2} (2u u_y + 2v v_y) \end{aligned}$$

from (A) & (B)

--- (B)

$$\left(\frac{\partial}{\partial x} |f(z)| \right)^2 + \left(\frac{\partial}{\partial y} |f(z)| \right)^2$$

$$= \frac{1}{4} (u^2 + v^2)^{-1} \left[(2u u_x + 2v v_x)^2 + (2u u_y + 2v v_y)^2 \right]$$

$$= \frac{1}{4(u^2 + v^2)} \left[4u^2 u_x^2 + 4v^2 v_x^2 + 8uv u_x v_x + 4u^2 u_y^2 + 4v^2 v_y^2 + 8uv u_y v_y \right]$$

$$u_x = v_y, \quad u_y = -v_x$$

$$= \frac{1}{4(u^2 + v^2)} [4u^2 u_x^2 + 4v^2 v_x^2 + 4u^2 v_x^2 + 4v^2 u_x^2 + 8uv u_x v_x - 8uv u_x v_x]$$

$$= \frac{1}{\cancel{4(u^2 + v^2)}} [\cancel{4(u^2 + v^2)} (u_x^2 + v_x^2)]$$

$$= u_x^2 + v_x^2$$

$$= |f'(z)|^2$$

$$\begin{aligned} f'(z) &= u_x + i v_x \\ |f'(z)| &= \sqrt{u_x^2 + v_x^2} \end{aligned}$$

$$\text{ie } \left(\frac{\partial}{\partial x} |f(z)| \right)^2 + \left(\frac{\partial}{\partial y} |f(z)| \right)^2 = |f'(z)|^2$$